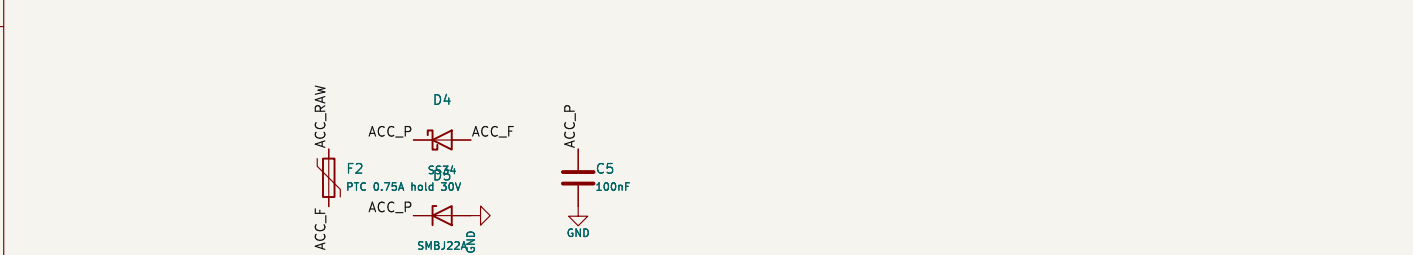
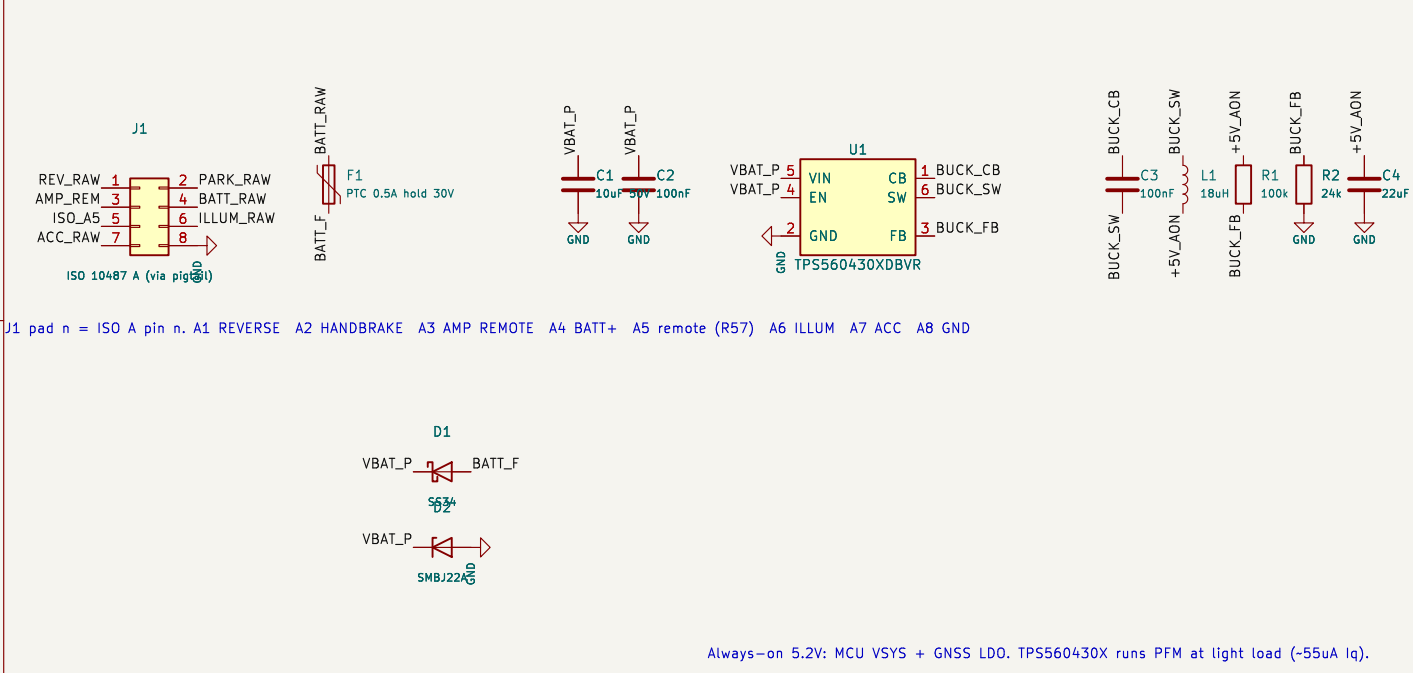
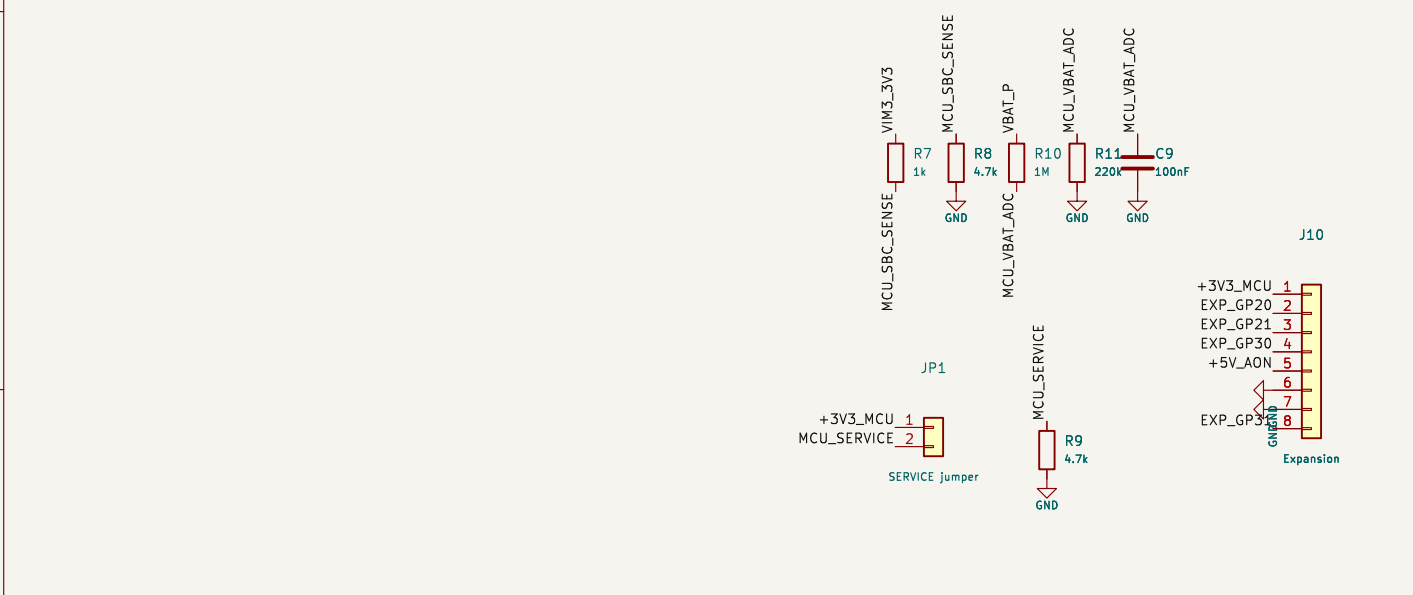


1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICCO AND GNSS

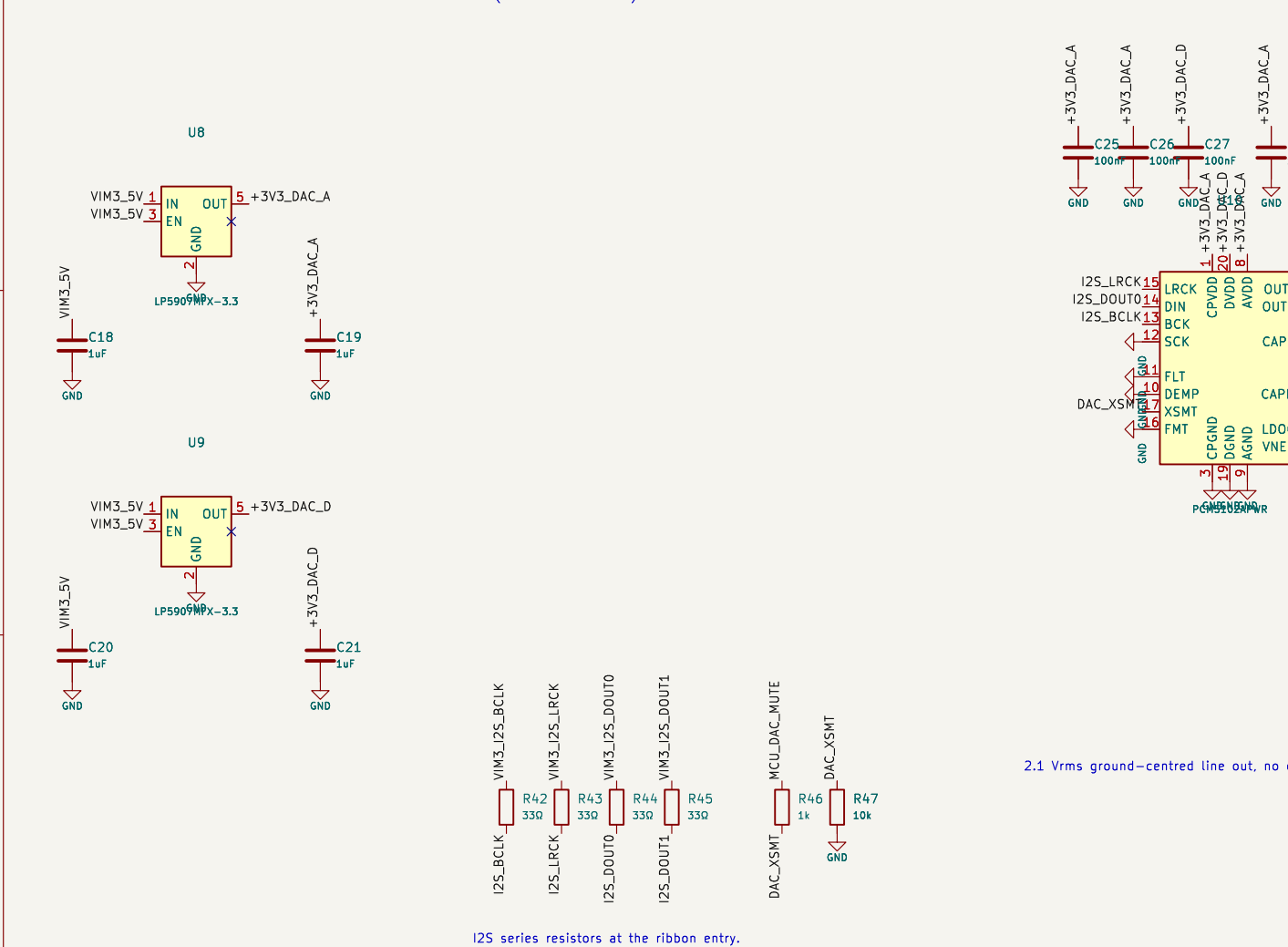


4. MCU HELPERS (MCU core is section 13)



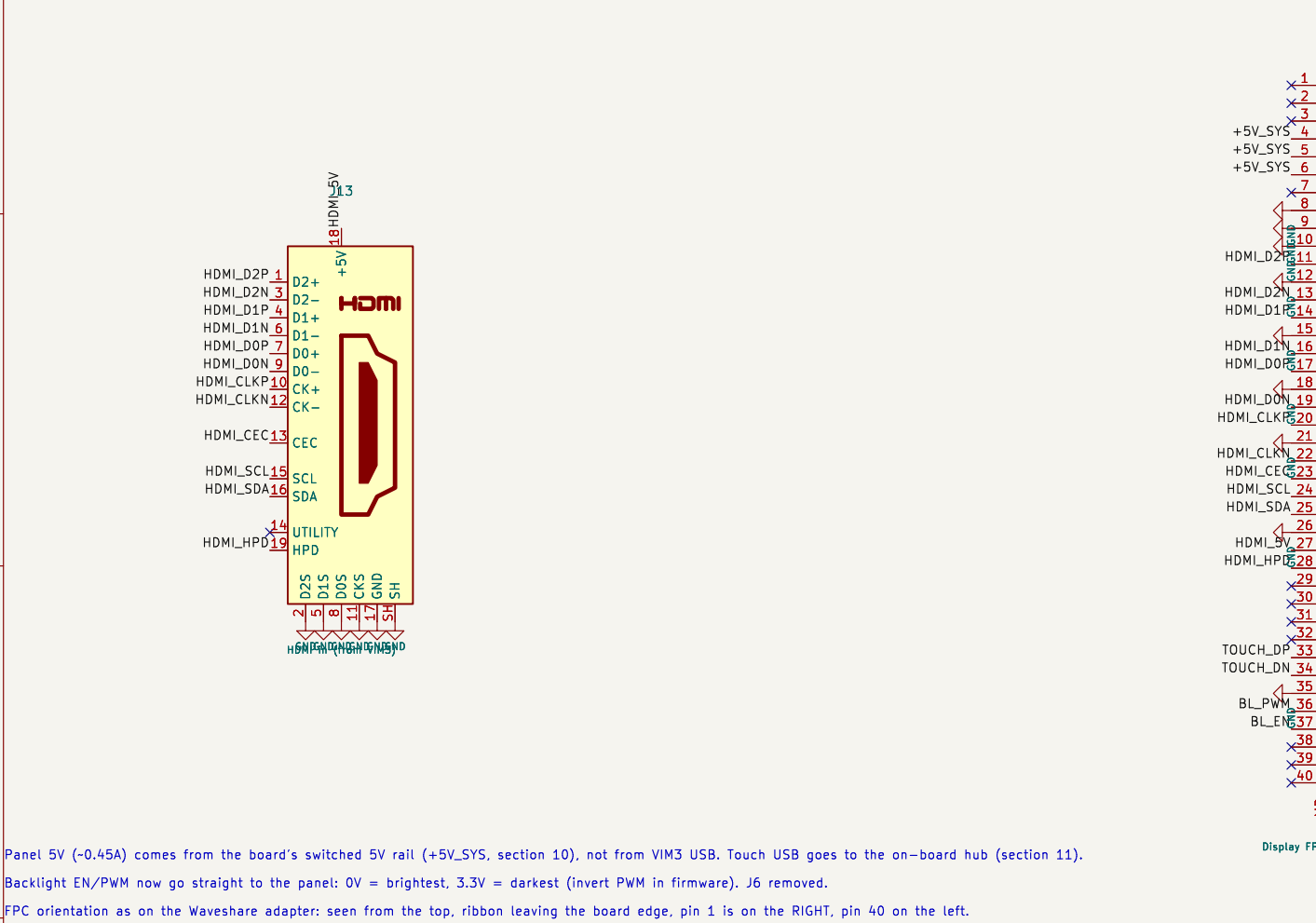
GPIO map see section 13 and DESIGN\_rev03.mxd

7. AUDIO: VM3 TDM-B I2S -> 2x PCM5102A (front + rear) -> line out on ISO C1



Car radio peripheral board rev 0.3 - MG F (1996), VM3 + RP2350B, 4ch line out, NEO-M9N. Generated: see DESIGN\_rev03.mxd.

8. DISPLAY: Waveshare Tin 70H-1024x600 adapter built in (HDMI in from VM3 -> A0-pin FPC to the panel)

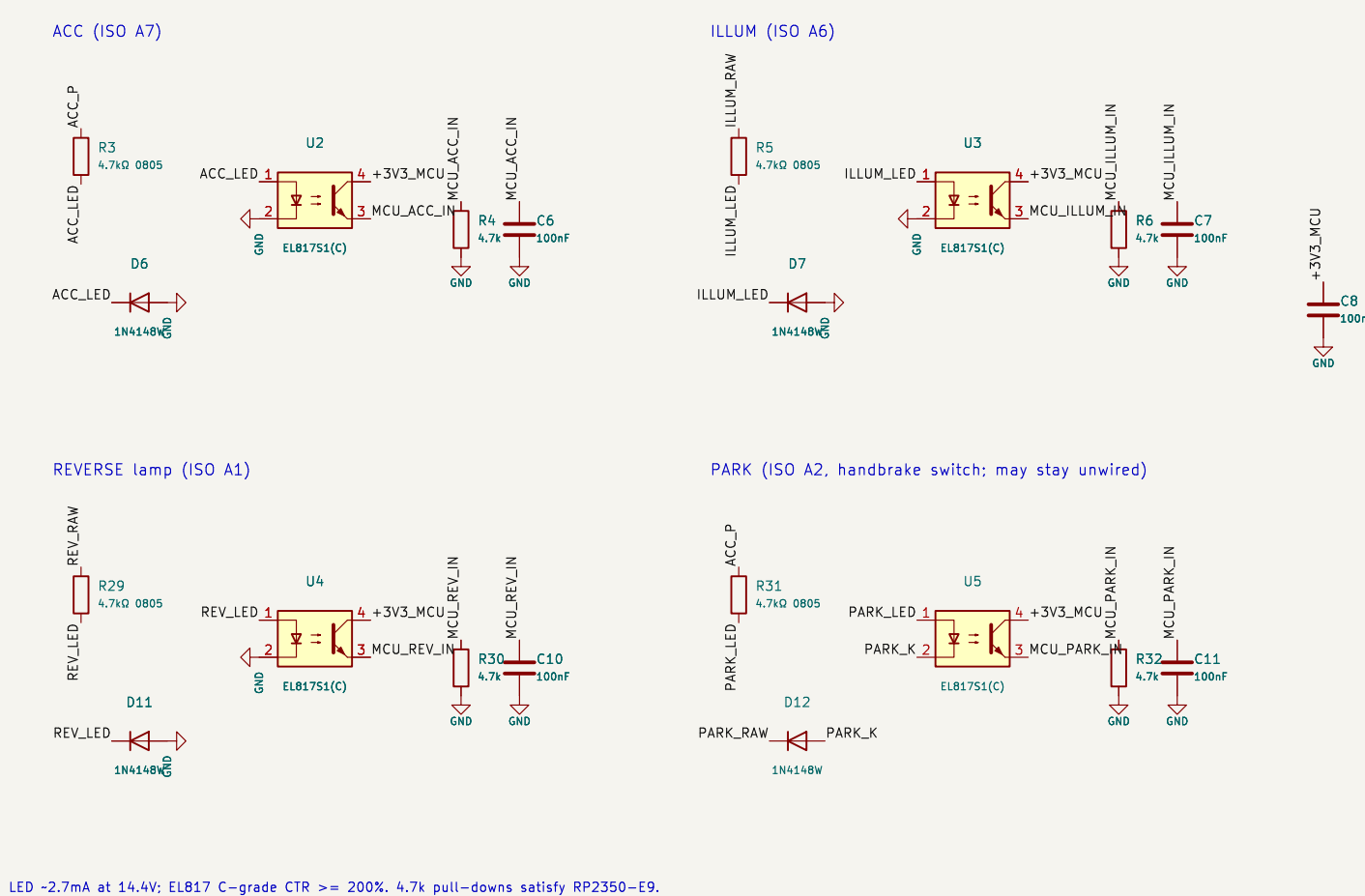


Panel 5V (4-6A) comes from the board's switched 5V rail (4-9.31V, section 10), not from VM3 USB. Touch USB goes to the on-board hub (section 11).

Backlight 15V/40mA max go through to the panel 5V -> Integrated, 5.2V -> external (from Panel in forward), JE reference.

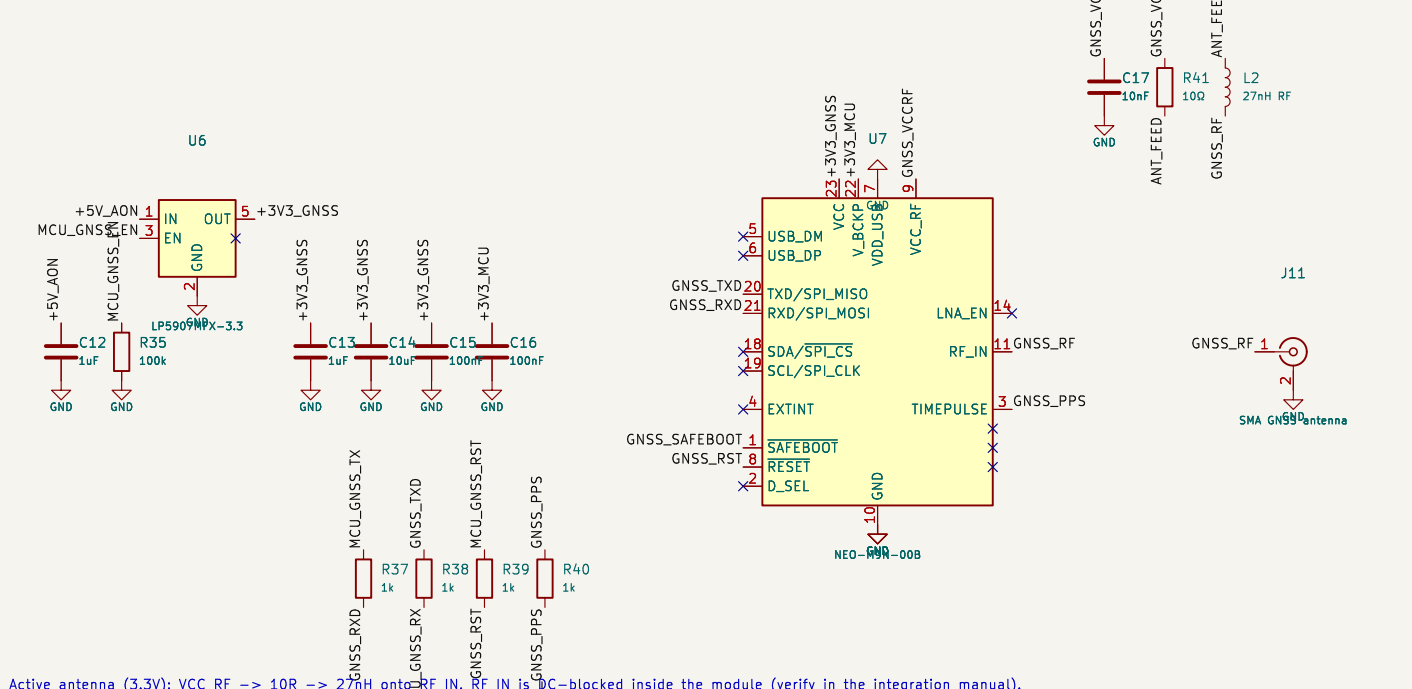
FPC orientation: as on the developer's adapter: seen from the top, ribbon facing the board edge, pin 1 is on the right, pin 40 on the left.

2. VEHICLE INPUTS (opto-isolated, ACC, ILLUM, REVERSE: +12V = on, PARK: switch to ground = on).



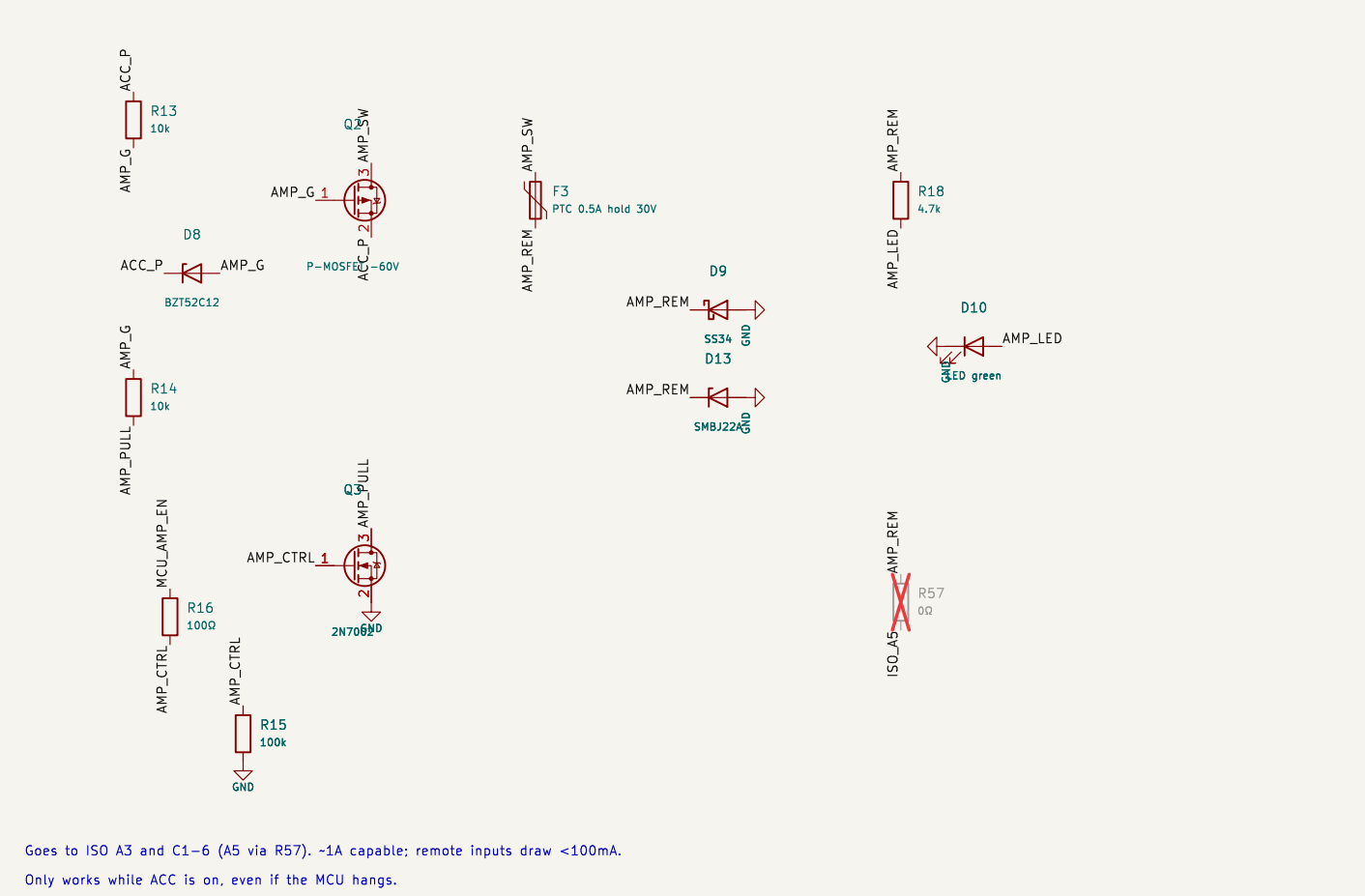
LED -2-7mA at 14-16V, 1.8V/1.8V C-grade CTR >= 200%, 4.7k pull-down safety RP2350-15.

5. GNSS: u-blox NEO-M9N, active antenna on SMA



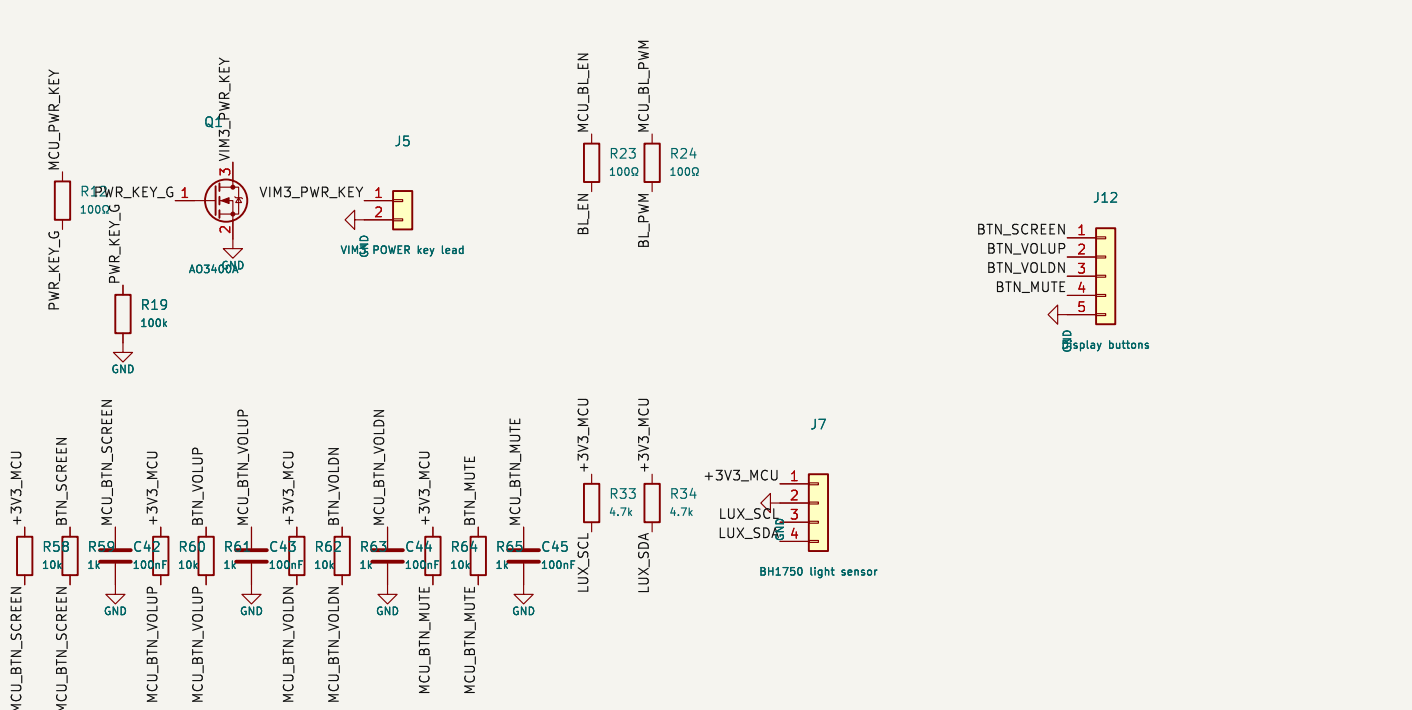
Active antenna (E30) VCC-RT -> I2C -> 27MHz amplifier, RTN -> shielded inside the module (entry in the integration manual).

3. AMPLIFIER REMOTE (+12V from ACC, MCU GPIO)



Open to ISO A5 and C4 +6-8 Ohm via RTN, +12V regulator remote supply draw <100mA. Only works with ACC is not, even if the MCU keeps.

6. VM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the MCU)



Display buttons (by GHS): GHS screen, GHS1, GHS2, GHS3, GHS4, GHS5, GHS6, GHS7, GHS8, GHS9, GHS10, GHS11, GHS12, GHS13, GHS14, GHS15, GHS16, GHS17, GHS18, GHS19, GHS20, GHS21, GHS22, GHS23, GHS24, GHS25, GHS26, GHS27, GHS28, GHS29, GHS30, GHS31, GHS32, GHS33, GHS34, GHS35, GHS36, GHS37, GHS38, GHS39, GHS40, GHS41, GHS42, GHS43, GHS44, GHS45, GHS46, GHS47, GHS48, GHS49, GHS50, GHS51, GHS52, GHS53, GHS54, GHS55, GHS56, GHS57, GHS58, GHS59, GHS60, GHS61, GHS62, GHS63, GHS64, GHS65, GHS66, GHS67, GHS68, GHS69, GHS70, GHS71, GHS72, GHS73, GHS74, GHS75, GHS76, GHS77, GHS78, GHS79, GHS80, GHS81, GHS82, GHS83, GHS84, GHS85, GHS86, GHS87, GHS88, GHS89, GHS90, GHS91, GHS92, GHS93, GHS94, GHS95, GHS96, GHS97, GHS98, GHS99, GHS100, GHS101, GHS102, GHS103, GHS104, GHS105, GHS106, GHS107, GHS108, GHS109, GHS110, GHS111, GHS112, GHS113, GHS114, GHS115, GHS116, GHS117, GHS118, GHS119, GHS120, GHS121, GHS122, GHS123, GHS124, GHS125, GHS126, GHS127, GHS128, GHS129, GHS130, GHS131, GHS132, GHS133, GHS134, GHS135, GHS136, GHS137, GHS138, GHS139, GHS140, GHS141, GHS142, GHS143, GHS144, GHS145, GHS146, GHS147, GHS148, GHS149, GHS150, GHS151, GHS152, GHS153, GHS154, GHS155, GHS156, GHS157, GHS158, GHS159, GHS160, GHS161, GHS162, GHS163, GHS164, GHS165, GHS166, GHS167, GHS168, GHS169, GHS170, GHS171, GHS172, GHS173, GHS174, GHS175, GHS176, GHS177, GHS178, GHS179, GHS180, GHS181, GHS182, GHS183, GHS184, GHS185, GHS186, GHS187, GHS188, GHS189, GHS190, GHS191, GHS192, GHS193, GHS194, GHS195, GHS196, GHS197, GHS198, GHS199, GHS200, GHS201, GHS202, GHS203, GHS204, GHS205, GHS206, GHS207, GHS208, GHS209, GHS210, GHS211, GHS212, GHS213, GHS214, GHS215, GHS216, GHS217, GHS218, GHS219, GHS220, GHS221, GHS222, GHS223, GHS224, GHS225, GHS226, GHS227, GHS228, GHS229, GHS230, GHS231, GHS232, GHS233, GHS234, GHS235, GHS236, GHS237, GHS238, GHS239, GHS240, GHS241, GHS242, GHS243, GHS244, GHS245, GHS246, GHS247, GHS248, GHS249, GHS250, GHS251, GHS252, GHS253, GHS254, GHS255, GHS256, GHS257, GHS258, 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